Ali Husain

SOFTWARE ENGINEER AND PRODUCT DESIGNER

I design and build complete systems: the interface people use, the data and services behind it, and the infrastructure that keeps it running. Each one is specified first, tested before it ships, and documented to run without me.

1,257	15	9	2
automated tests, run in CI on every change	services run in production, 24/7	open-source projects, each documented and released	live offline-first apps, no account and no server

AREAS OF EXPERTISE

AI & LLM engineering
Agents, multi-agent workflows, LLM evaluation, prompt engineering, local models

Cloud, DevOps & SRE
Linux, containers, CI/CD, infrastructure as code, scheduled automation, monitoring, incident response

Networking & security
DNS, TLS, mesh VPN, firewalls, intrusion prevention, encryption, secrets handling

Backup & disaster recovery
Verified backups, multi-site copies, automated restore drills, rebuild runbooks

Software & web products
Python, TypeScript, Go, C/C++, Bash, SQL: offline-first web apps, services, command-line tools, technical writing

Generative media
Prompt systems and frame QA for image, video and music models; editing, sound design, loudness-checked 4K delivery

EXPERIENCE

AI Systems Engineer

`github / evalbench · aih · agent-guardrails`

Independent · agents, evaluation and model deployment

- **Built Evalbench, a self-hosted LLM evaluation workbench:** blind pairwise judging, seven automatic check types including code tests in a sandbox with no network, and regression tracking across local and API models; 107 automated tests.
- **Designed and run an autonomous AI operations agent** that monitors and maintains production infrastructure around the clock: bound by a written charter, woken by alerts, every change logged.
- **Built multi-agent delivery pipelines** of up to 9 coordinated agents (1.45M tokens, 566 tool calls) with build, critique and automated browser-verification stages, and ran 13-agent system audits.
- **Built a knowledge pipeline and agent guardrails:** 4 GB+ of conversations from three machines in one full-text archive, LLM extraction cross-checked by a second model, and a hook that screens agent commands before they run.

Infrastructure & DevOps Engineer

`github / homelab · cockpit`

Independent · self-hosted production platform, run as code

- **Operate a 15-service containerized platform** (14 health-checked) behind one TLS edge: DNS, password and identity management, file sync, smart-home control; remote access over a 7-node mesh VPN with zero port forwards.
- **Built Cockpit, a read-only operations dashboard:** one 8 MB Go binary with a TypeScript UI fed by Server-Sent Events, run as privilege-split systemd services; 113 automated tests, including a 33-check end-to-end suite under real systemd.
- **Designed backup & disaster recovery:** 12 integrity-verified sets nightly, 3 copies on 3 machines, automated weekly restore drills, a 2–3 hour bare-metal rebuild plan.
- **Built observability for 77 failure conditions** plus 13 uptime probes at 60 s, self-healing watchdogs and a dead-man switch; everything as code: 376 commits, 20 scheduled services, drift detection, ~3,800 lines of runbooks and notes.

CONTACT

EMAIL
ali@gallowforge.dev

PORTFOLIO
gallowforge.github.io

GITHUB
github.com/gallowforge

LOCATION
Remote · UTC+3

TOOLBOX

AI platforms
Anthropic and OpenAI models, open-weight LLMs, local inference on GPU, agent frameworks, MCP, Hugging Face ecosystem

Generative media
Image, video, voice and music generation models; diffusion pipelines; programmatic editing with Python and ffmpeg

Infrastructure
Linux (Debian, Arch), Docker, systemd, Git, GitHub Actions, Caddy, Nginx, Tailscale / WireGuard, SQLite, Redis, CouchDB

Web & product
Astro, React, Preact, Tailwind, FastAPI, PWAs, Playwright, Vitest, Lighthouse CI, accessibility (WCAG)

Languages
Python, TypeScript / JavaScript, Go, C, C++, Bash, SQL, R, Standard ML, Prolog

OPEN TO

- **Remote contracts:** web products, infrastructure, DevOps, automation, AI integration, video and audio post-production.
- **AI training & evaluation:** code review, model-output rating, prompt and rubric writing.
- **Flexible hours:** UTC+3 overlaps the full European working day.

Web & Product Developer

github / Ethereal · ledger

Independent · product design, front-end and delivery

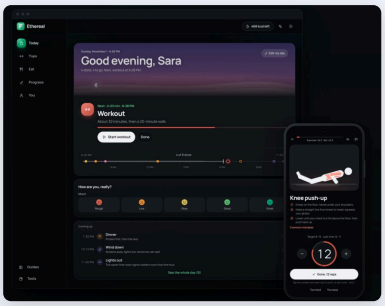
Ethereal

LIVE

Offline-first health & fitness app · product lead

Led product design and delivery of an installable, account-free app that plans the day, coaches workouts and tracks nutrition. 41 animated exercises, a 1,474-food offline database and 392 automated tests, built to WCAG 2.1 AA and Lighthouse 100. gallowforge.github.io/Ethereal

TypeScript · Astro · PWA · CI/CD · UX design



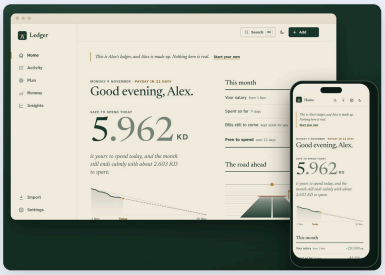
Ledger

LIVE

Offline-first personal finance app · product lead

Turns savings into a monthly salary that lasts: what is safe to spend today and how long the money runs. Reads bank statements on the device and reconciles every row against the running balance. No account, no server; 483 automated tests. gallowforge.github.io/ledger

TypeScript · Astro · Preact · pdf.js · PWA



Sites, 3D and 40+ interactive tools

A six-page animated brand site for an AI video-ad studio; a scroll-driven WebGL story page verified by automated screenshots at phone and desktop sizes; live market dashboards on real-time price feeds, learning tools, data visualizations and planners, each designed for phone and desktop.

Next.js · Tailwind · Framer Motion · WebGL · Puppeteer · real-time data

Video, Audio & Generative Media Producer

github / filmforge

Independent · production method, editing, sound and delivery

- **Built filmforge, a documented production method** for painted films made with AI image and video models: nine Claude Code skills (story, style lock, prompt grammar, frame QA, motion, sound, delivery) and seven tested Python tools that price, lint and verify the work at every stage gate.
- **Post-production built in code:** beat-matched editing, color grading, motion graphics and animated titles, narration timed across cuts, layered sound design and mixing to loudness targets, 4K mastering on GPU.
- **Delivery checked, not assumed:** every encode verified for streams, loudness and picture-to-music sync, and each production archived with a SHA-256 completeness proof anyone can re-verify.

Open-Source Tools

github / agent-guardrails · aih · bwtidy

Independent · small tools, each tested and released

- **agent-guardrails:** reviewer and verifier agents that judge work they did not write, and a hook that warns a coding agent before a destructive shell command runs; 36 automated tests.
- **aih:** every AI conversation from three machines in one local full-text index, searchable from the command line; 31 automated checks.
- **bwtidy:** audits and tidies a password vault entirely on the device, and changes nothing until you approve the plan; 12 automated tests.

TESTED IN PUBLIC

ledger	483
Ethereal	392
cockpit	113
evalbench	107
homelab	43
filmforge	40
agent-guardrails	36
aih	31
bwtidy	12

Automated tests per repository, run in CI on every change: 1,257 in all.

HOW I WORK

- **Prove it works.** A backup isn't done until it's restored; a fix isn't done until it's tested from the user's side.
- **Write it down.** Every system ships with a runbook someone else could follow.
- **Fewer things, done properly.** Simple setups that stay up beat clever ones that break.
- **Learn by shipping.** Linux, Go, AI agents, post-production: each learned by building something real.

EDUCATION

Computer Science and Chemical Engineering, Kuwait University

Continuous self-study

Algorithms, computer science and AI, each put to use in a shipped project

BEYOND WORK

Cinema (1,150+ films logged) and storytelling.

Scan for code and live work